

Blind Disturbance-Model Selection for Diffusion-Aided Semantic Image Reception

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Abstract—A receiver with a learned interference prior can either separate an interfering transmission or absorb it into an effective noise model. Always invoking the more detailed model need not yield the most faithful reconstruction when interference power is unknown and inference is approximate. We study this decision for a frozen semantic image codec and introduce a blind selector between a Gaussian diffusion denoiser and a calibrated joint signal–interference sampler. The selector uses received energy and, when supported by development validation, block-energy heterogeneity; it evaluates only the selected receiver. A shared-checkpoint protocol separates receiver choice from retraining and compares the method with a blind total-disturbance denoiser, fixed interference cancellation, and the same cancellation receiver with a direct-decoding gate. On 256 independent image pairs with two random seeds and 18 channel conditions, the frozen rule attains 21.593 dB mean PSNR: +1.334 dB relative to calibrated joint sampling and +0.750 dB relative to the blind Gaussian receiver. It uses 52.2 total predictor evaluations per frame on average, compared with 80 for joint sampling. The scope is image reconstruction under the specified interference family, rather than a guarantee of semantic correctness or universal channel robustness.

Index Terms—Semantic communications, interference cancellation, diffusion models, receiver adaptation, deep joint source–channel coding.

I. INTRODUCTION

Deep joint source–channel coding maps an image directly to channel symbols, allowing the learned decoder to accommodate transmission noise [1], [2]. A frozen codec, however, can encounter co-channel transmissions that differ substantially from its training noise. A receiver-side generative prior offers two useful responses. It may treat the aggregate disturbance as Gaussian and denoise the received latent, as motivated by channel denoising diffusion models (CDDM) [3]. Alternatively, it may represent both desired and interfering latents and infer them jointly, as in interference cancellation diffusion models (ICDM) [4].

These choices impose different assumptions and computational costs. A Gaussian approximation discards interference structure, but avoids estimating a second latent and its unknown amplitude. Explicit separation preserves a structured interference hypothesis, but must resolve an underdetermined mixture using imperfect priors and finite-step inference. Even accurate measurement fitting does not establish that the desired component has been recovered. Thus, adding an interference prior is not by itself a reason to invoke it for every frame.

This leads to our question: *can a receiver choose between these two disturbance models, without observing the clean image or true interference power, more effectively than it can*

commit to either model? The relevant comparator is a Gaussian receiver that estimates the *total* disturbance. Comparing only against a denoiser calibrated to thermal noise would mix the benefit of power estimation with the benefit of explicit separation. Likewise, a new cancellation scheme with a bypass must be compared with an existing cancellation scheme using the same bypass.

We introduce a small, interpretable selector operating before generative inference. It uses a global excess-energy statistic and an optional block-energy heterogeneity statistic, selects one receiver, and decodes once. All neural weights remain fixed. The contribution is this receiver decision and its controlled evaluation, not a new diffusion prior or a claim to be the first blind generative receiver. Specifically, we provide:

- a receiver-only formulation separating Gaussian disturbance approximation from explicit interference reconstruction;
- a development-fitted selection rule with image-group validation and no test-time clean-image access;
- paired independent confirmation, bypass controls, and mismatched-block tests reporting reconstruction quality, harmful outcomes, and inference cost.

II. RELATED WORK AND SCOPE

CDDM matches channel noise to a diffusion noise level and includes a three-stage training procedure with decoder adaptation [3]. ICDM instead uses desired-signal and interference priors for joint inference [4]. We adapt these receiver ideas to shared frozen weights to isolate the disturbance-model decision. Consequently, our Gaussian receiver is a receiver-only port, not a reproduction of the complete CDDM system; our results cannot establish superiority over its separately trained codec.

Receiver adaptation also extends beyond these two methods. ISEC iteratively corrects latent errors using a learned source prior [5]. DiffCom uses channel observations to guide generative reconstruction and includes blind inference of unknown transmission characteristics [6]. Blind inverse-problem methods such as GibbsDDRM estimate unknown forward-model parameters during diffusion inference [7]. These works establish that blind generative inference and receiver-side adaptation are existing ideas. Our narrower question concerns selecting between two already available disturbance treatments before their inference cost is incurred. We do not infer a posterior model probability from the selector score, nor interpret residual reduction as a semantic certificate.

III. SYSTEM MODEL AND RECEIVER CHOICE

A. An Interfering Latent Transmission

Let $\mathbf{u} \in [0, 1]^d$ be the desired image, and let the frozen encoder produce $\mathbf{x} \in \mathbb{C}^n$. A separate transmitter produces interfering symbols $\mathbf{z} \in \mathbb{C}^n$. Each transmitter normalizes its frame to unit average complex symbol power. The additive channel is

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{w}_b, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad a_b \geq 0, \quad (1)$$

where b indexes contiguous symbol blocks. The channel bandwidth ratio is n/d . The receiver knows N_0 , the normalization convention, and a chosen block partition, but not a_b , the interference profile, $\mathbf{z}$, or $\mathbf{u}$. The test channel has unit desired-channel gain. Fading, channel estimation, and synchronization are outside this experiment.

For a prescribed nominal SINR γ , the simulator sets average nominal interference power to $10^{-\gamma/10} - N_0$. The instantaneous interference energy also depends on the realized block energies of $\mathbf{z}$; nominal SINR must therefore not be confused with measured per-frame SINR.

The explicit model associates two learned priors with (1):

$$p(\mathbf{x}, \mathbf{z} \mid \mathbf{y}, \mathbf{a}) \propto p_s(\mathbf{x}) p_z(\mathbf{z}) \times \exp \left[-\frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|^2 \right]. \quad (2)$$

This equation describes the model; our finite-step sampler is an approximation, not an exact posterior draw or a proven MAP optimizer. Without priors, the change $\mathbf{x}'_b = \mathbf{x}_b + a_b \delta_b$, $\mathbf{z}'_b = \mathbf{z}_b - \delta_b$ leaves the mixture unchanged. Power constraints remove some ambiguity but do not generally identify both latents. The Gaussian model instead uses

$$p_G(\mathbf{y} \mid \mathbf{x}; \nu) = \mathcal{CN}(\mathbf{y}; \mathbf{x}, \nu \mathbf{I}), \quad \nu \geq N_0. \quad (3)$$

It sacrifices interference structure to avoid explicit component estimation.

B. Two Frozen Candidate Receivers

For Gaussian reception, estimate the disturbance from raw received symbols:

$$e_b = \frac{\|\mathbf{y}_b\|^2}{L_b}, \quad \bar{e} = \frac{1}{n} \sum_b L_b e_b, \quad (4)$$

$$\hat{P} = [\bar{e} - 1 - N_0]_+, \quad \hat{\nu} = N_0 + \hat{P}.$$

With unit source power and vanishing signal–interference cross expectation, unclipped excess energy estimates mean interference energy. Finite blocks, latent correlations, and clipping prevent treating it as an exact amplitude estimate. In particular, local desired-signal energy can resemble interference heterogeneity.

Let $r(\mathbf{y}) \in \mathbb{R}^{2n}$ stack real and imaginary components. Given the training diffusion schedule $\bar{\alpha}_t$, the Gaussian receiver G starts at the nearest noise level satisfying

$$\frac{1 - \bar{\alpha}_{t_0}}{\bar{\alpha}_{t_0}} \simeq \frac{\hat{\nu}}{2}, \quad \mathbf{v}_{t_0} = \sqrt{\bar{\alpha}_{t_0}} r(\mathbf{y}). \quad (5)$$

The factor $1/2$ converts complex noise power to variance per real coordinate. For decreasing respaced indices, a noise predictor ϵ_s gives

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{v}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_s(\mathbf{v}_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (6)$$

$$\mathbf{v}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \epsilon_s(\mathbf{v}_t, t). \quad (7)$$

At most 40 predictor evaluations are used, with fewer when the matched initial index is below 39. A common normalization and the frozen decoder produce the reconstructed image.

The structured receiver S initializes $a_b^{(0)} = \sqrt{[e_b - 1 - N_0]_+}$ and performs joint diffusion inference with signal and interference predictors. During late corrector updates it obtains denoised, frame-normalized estimates $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$, then updates

$$a_b^+ = \left[\frac{\Re\{\tilde{\mathbf{z}}_b^H (\mathbf{y}_b - \tilde{\mathbf{x}}_b)\}}{\|\tilde{\mathbf{z}}_b\|^2} \right]_+. \quad (8)$$

An empty denominator retains the previous amplitude. This is a nonnegative least-squares update conditional on point estimates, not a correction for their uncertainty. Calibration begins when the signal coefficient $\alpha(t) \geq 0.5$. Estimated block SINR updates the inherited guidance coefficients by clipped linear interpolation of the original AWGN table. We use the existing 40-step, order-three joint sampler: 40 signal and 40 interference evaluations. The initialization-only variant keeps amplitudes fixed and supplies an ablation of calibration.

C. Selection Before Generative Inference

The selector augments $\hat{P}$ with normalized block-energy variation

$$H = \frac{B^{-1} \sum_{b=1}^B (e_b - \bar{e})^2}{\max(\bar{e}^2, \epsilon)}. \quad (9)$$

All experimental blocks have equal length. This statistic describes received energy variation, not an identified interference distribution. For $\theta = (\tau_\ell, \tau_h, \eta)$, define

$$\pi_\theta(\mathbf{y}) = \begin{cases} \text{G}, & \hat{P} \leq \tau_\ell, H \leq \eta, \\ \text{G}, & \hat{P} \leq \tau_h, H > \eta, \\ \text{S}, & \text{otherwise.} \end{cases} \quad (10)$$

We also test the reversed assignment of G and S in (10); the Gaussian branch is not assumed to belong to the low-power region. The energy-only family sets $\tau_\ell = \tau_h$. Either family can reduce to a constant receiver; heterogeneity need not be useful.

We fit thresholds to mean per-image PSNR on development data. Eight folds keep all channel conditions belonging to one image together. Each fold is predicted using thresholds fitted on the other seven, then the family with the highest out-of-fold mean is refitted on all development images. Ties prefer lower component inference cost and then the simpler family. If neither family improves on the better fixed receiver in development validation, that fixed receiver is retained. No threshold is adjusted using confirmation or pressure results. At reception, (10) is computed first and only its chosen branch

is executed. Selection adds $O(n)$ energy calculations and no diffusion evaluations.

For analysis, let $q_j(\mathbf{u}, \mathbf{y})$ be a candidate's PSNR. The unavailable per-frame oracle has quality $\max(q_G, q_S)$. The exact identity

$$\max(q_G, q_S) - q_{\pi_\theta} = |q_G - q_S| \mathbf{1}\{\pi_\theta \neq \arg \max_j q_j\} \quad (11)$$

shows that complementary receivers are necessary but not sufficient: mistakes on large-gap frames matter most. This is an accounting identity, not a generalization bound. Mean PSNR is the fitting objective; neither mean MSE, LPIPS nor harmful-tail probability is implicitly optimized.

IV. EXPERIMENTAL EVALUATION

A. Protocol and Strong Controls

The desired images are 128×128 center crops from CelebA [8]. CIFAR-10 interference images are cropped at 32×32 and nearest-neighbor upsampled to the encoder input size. The original training sets contain 162,770 desired and 50,000 interference images. The inherited codec uses three Swin stages with widths (128, 192, 256) and depths (2, 2, 6); its 8-channel 16×16 real latent corresponds to 1,024 complex symbols and CBR 1/48. Five frozen checkpoints contain the desired encoder/decoder, interference encoder, and two DiT predictors, each with 28 blocks, width 1,152, 16 heads and patch size two. The original training schedule uses 100 interference-codec epochs, 40 desired-codec epochs, and 50/20 epochs for the signal/interference predictors. This study adds no neural retraining.

Development uses 64 validation image pairs and seed 1700. Confirmation reserves 256 previously unevaluated test pairs with seeds 2700 and 2701. SNR is 20 dB; nominal SINR is $-7, -4, 0, 4, 7, 10$ dB. Sixteen blocks of 64 complex symbols follow stationary, alternating (0.25, 1.75), or burst (0, 2) relative-power patterns. Each pair is evaluated under all 18 conditions. The confirmation set therefore has 9,216 received frames but only 256 image-pair clusters. Confirmation starts only after policy fitting ends. File-hash audits exclude overlap with development, pressure images and the first 200 test pairs used in earlier project evaluations. The independence claim is at image-pair level, not at person-identity level.

We compare direct decoding; the same Gaussian port using only N_0 ; the blind-total-disturbance Gaussian receiver G; initialization-only joint sampling; calibrated joint sampling S; and the online selector. A same-gate control decodes directly when $\hat{P} \leq 0.15$ and otherwise uses S. Its output quality is reconstructed exactly from paired branches; its reported cost excludes gate overhead. Development additionally examines the previous coupled-feedback receiver and a Gaussian port given nominal total disturbance power. The latter is a privileged diagnostic, not a guaranteed upper bound. Neither diagnostic is used as a confirmatory claim of superiority over a published system.

The prespecified power thresholds are 0, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, 8 and 10^6 ; heterogeneity thresholds are

$\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$. Including both assignment directions, the families contain 26 and 2,028 candidate rules. All methods see identical received tensors. Channel randomness is separate from sampler randomness; both CPU and GPU sampler states are reset per frame. Measurements use float32, batch one, TF32 disabled, four CPU threads, and one RTX 4090. Synchronized receiver time includes selection, sampling and decoding, but excludes encoding, image-quality evaluation and one-time model loading. Signal and interference model calls are counted separately.

We report arithmetic means of per-image PSNR and MSE, LPIPS-VGG v0.1 [9], and four-scale MS-SSIM [10], always clipping RGB to $[0, 1]$. Four scales fit the image resolution; these values are not compared with five-scale scores. Paired 95% intervals resample image-pair clusters 5,000 times while retaining both seeds and all conditions. Harm means a PSNR loss exceeding 0.5 dB relative to a specified comparator. These descriptive intervals do not certify safety over arbitrary future channels.

B. Does Receiver Choice Generalize?

Image-group validation yields 22.002 dB for energy-only selection and 22.239 dB for the two-feature family. The retained energy and heterogeneity rule uses $\tau_\ell = 8$, $\tau_h = 0.5$ and $\eta = 0.2$, with Gaussian reception below the applicable power threshold. In the low-heterogeneity development region (810 frames), the largest observed excess-energy estimate is 5.241; the fitted threshold 8 therefore lies beyond observed support and acts as an always-Gaussian rule in that region. It is not an identified physical transition, and the family comparison does not establish that either feature is individually necessary. The earlier coupled-feedback receiver obtains 20.477 dB, versus 20.605 dB for calibrated joint sampling and 20.835 dB for its same-gate control. The privileged nominal-power Gaussian diagnostic obtains 21.591 dB. These development comparisons motivate the receiver decision but are not independent confirmation. Table I and Fig. 1 report the independent experiment. The selector changes mean PSNR by +1.334 dB relative to calibrated joint sampling (95% interval [1.265, 1.404]), +0.750 dB relative to blind Gaussian denoising (95% interval [0.698, 0.806]), and +1.140 dB relative to the same-gate control (95% interval [1.075, 1.205]). Both fixed-receiver comparisons have positive paired intervals, supporting a mean-PSNR benefit within the matched confirmation population. Estimating total disturbance changes the Gaussian receiver from 18.829 to 20.843 dB, so the thermal-noise-only port is insufficient as the sole Gaussian baseline. Disaggregating blind Gaussian/joint/selector PSNR gives stationary 21.310/19.282/21.317 dB with 98.8% Gaussian choices, alternating 20.737/20.178/21.285 dB with 59.5% Gaussian choices, burst 20.481/21.317/22.177 dB with 50.1% Gaussian choices. This profile dependence is consistent with complementary receiver utility, but does not identify the physical cause of the heterogeneity statistic.

TABLE I
INDEPENDENT CONFIRMATION. QUALITY IS AVERAGED PER IMAGE AND FRAME. NFE COUNTS BOTH PRIORS; LATENCY IS SYNCHRONIZED BATCH-ONE RECEIVER TIME. THE SAME-GATE CONTROL’S LATENCY IS A COMPONENT ESTIMATE.

Receiver	PSNR	MSE $\times 10^3$	LPIPS	MS-SSIM	NFE	ms
Direct	18.67	29.14	0.440	0.697	0.0	8
Gaussian (N0)	18.83	28.70	0.438	0.700	18.0	225
Blind Gaussian	20.84	24.90	0.378	0.782	40.0	488
Init. joint	19.61	20.15	0.363	0.793	80.0	1110
Calibrated joint	20.26	18.49	0.362	0.802	80.0	1180
Joint + gate	20.45	18.43	0.361	0.801	65.9	974
Selector	21.59	16.45	0.336	0.832	52.2	701

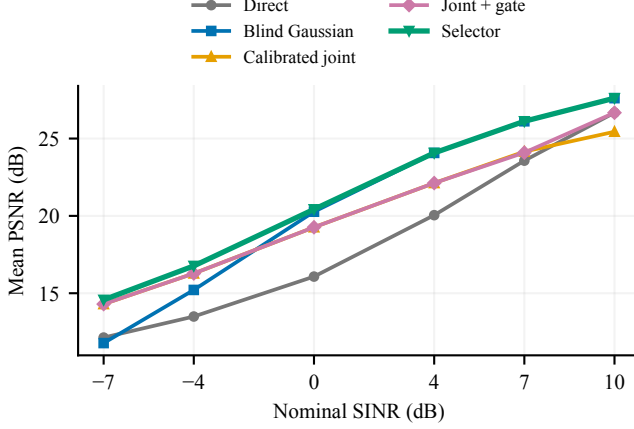


Fig. 1. Independent confirmation by nominal SINR, averaging all three interference profiles. Curves distinguish Gaussian denoising, explicit separation and their frozen selection; they do not compare separately retrained published systems.

C. What Is the Cost of a Wrong Decision?

The selector chooses Gaussian reception on 69.5% of confirmation frames. Its average cost is 52.2 predictor evaluations and 701 ms, versus 80 evaluations and 1180 ms for joint sampling. Against that receiver, 3.79% of frames lose more than 0.5 dB, and the fifth percentile of the paired change is -0.127 dB (Fig. 2). The unavailable per-frame oracle reaches 21.760 dB; selection leaves 0.167 dB mean oracle regret. Mean MSE changes from 0.01849 to 0.01645, LPIPS from 0.3618 to 0.3362, and MS-SSIM from 0.8015 to 0.8318. These secondary metrics and the harmful tail must accompany the average PSNR comparison.

D. Block Mismatch and No Interference

Pressure tests use 64 further test pairs and seed 3700, with nominal SINR $-4, 4, 10$ dB. One profile shifts burst boundaries by half a receiver block; another draws segment lengths uniformly from 16 to 128 symbols and normalizes the binary on/off amplitude pattern. A separate SNR-20 condition has no interference. The receiver keeps its original block partition and frozen rule. These families are reported separately from matched confirmation. Table II separates the three pressure families. For shifted blocks, the selector changes

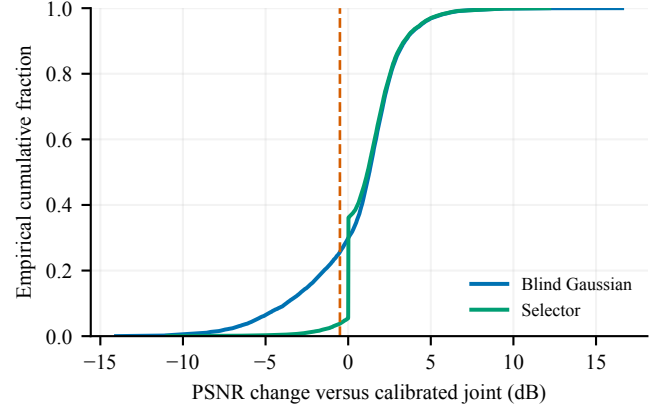


Fig. 2. Paired PSNR change on independent confirmation relative to calibrated joint sampling. Each point in the empirical CDF represents one received frame; uncertainty calculations cluster by image pair. The vertical line marks the predefined harmful-loss threshold.

TABLE II
PRESSURE TESTS. MEAN PSNR FOR EACH FAMILY. THE NO-INTERFERENCE ROW HAS SNR 20 dB; THE OTHER ROWS AVERAGE NOMINAL SINR $-4, 4, 10$ dB.

Profile	Direct	Gaussian	Joint	Select
Shifted blocks	20.83	22.99	21.09	23.00
Random lengths	20.64	23.42	22.74	23.98
No interference	31.37	31.37	27.73	31.37

mean PSNR by +1.914 dB relative to joint sampling and +0.010 dB relative to Gaussian denoising. For random lengths, the selector changes mean PSNR by +1.243 dB relative to joint sampling and +0.563 dB relative to Gaussian denoising. For no interference, the selector changes mean PSNR by +3.636 dB relative to joint sampling and +0.000 dB relative to Gaussian denoising. These shifted populations are diagnostic tests; no rule is refitted and their averages are not pooled with matched confirmation.

V. DISCUSSION AND CONCLUSION

The frozen blind selector improves mean PSNR over both fixed receiver choices in the matched independent study. It also improves on the same-gate control. Thus receiver-model choice earns a bounded empirical role, while harmful frames and mismatch results limit its deployment claim. The experiment isolates a receiver decision under a frozen codec and fixed source/interference priors. A different encoder, source distribution, SNR, interference population or inference budget can change the decision boundary. Block-energy heterogeneity can be caused by the desired latent itself, and the amplitude update uses uncertain point estimates. These facts limit any physical interpretation of the selected label. A branch choice is a prediction of reconstruction utility, not proof that the disturbance is Gaussian or that an interfering user has been identified.

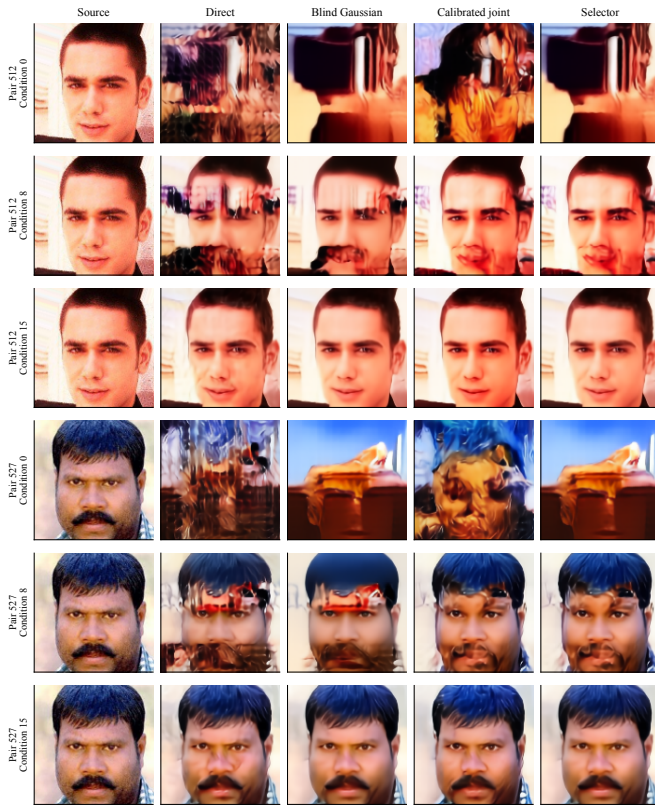


Fig. 3. All six qualitative examples reserved before confirmation: two image pairs crossed with conditions 0, 8 and 15. The gallery is exhaustive for the prespecified set and is not selected from observed reconstruction quality.

PSNR and MSE are different aggregations of the same pixel error, so an increase in one need not imply an improvement in the other. LPIPS and MS-SSIM provide additional reconstruction evidence but do not measure correctness of a driving, recognition or control task. Extending to such tasks requires task-specific losses, realistic fading and synchronization, and new validation data. Within the present scope, the practical question is whether a cheap blind decision earns its additional complexity over strong fixed receivers. The reported controls

and independent evaluation are designed to answer precisely that question.

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